

When Candidate-Dependent Context Can Leave a Common-Context Robot-Policy Winner Unidentified

Exact regret, graph repair, and finite-sample decisions

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Abstract

Pairwise robot-policy evaluations may report a global ranking even when each pair is tested in a different physical setting. Even with every pair observed, pair-specific contexts can admit opposite common-context winners. Under primitive pair responses, the constructed full-edge ambiguity set has a unique uniform minimax lottery, with regret $(K - 1)/(4K)$. A shared-binary-success sensitivity preserves that lottery and value, but the ambiguity then comes from missing target-context support rather than candidate-dependent routing.

In the shared-success model, identifying all within-context policy differences requires $m - 1$ comparable bridge types when the route graph has m components and the allowable bridge graph is connected. Repeating existing routes improves precision, not connectivity.

Simultaneous edge intervals propagate through the same graph constraints to target-difference bounds, possible and certified winners, and a minimax action by linear programming. Under an exact Bradley-Terry target, the winner set is identified exactly when the common-context comparison graph is connected.

Three pinned records show the operational distinction. Complete common-state panels identify exact finite-panel winners in three Ankle tasks. RoboArena’s pair-support graph remains incomplete and lacks a common-context bridge, while TRI’s released outcome tables omit the trial keys needed to reconstruct its published matched-bundle protocol.

Practitioner summary

- If the evaluated pair can change the task or scene, a precise pairwise leaderboard may still fail to determine which single policy is best for one common deployment population.
- The route graph diagnoses the structural gap: disconnected policy groups need safe, comparable cross-group comparisons in the same declared context.
- In the shared-success model, $m - 1$ bridge types identify all within-context differences across m components when allowable bridges connect them.
- With finite samples, use jointly valid edge intervals; the graph program inherits their coverage but cannot repair invalid sampling units.
- Connectivity does not ensure precision, validity, or safety.

1. The decision precedes the score

A robot-policy comparison can support several distinct actions:

1. choose one policy for a common deployment population;
2. choose between two policies presented under the same pair-specific mechanism;

3. make a same-mechanism choice within each future presented pair; or
4. report a mechanism-specific tournament score.

These actions are not interchangeable. A complete table of pairwise win probabilities identifies a tournament under the mechanisms that generated its edges. It need not identify how the policies compare in a common physical context population. The distinction matters when the evaluated pair can change the task, scene, operator behavior, or other context.

We distinguish identification, minimax decisions under ambiguity, and evidence connecting the target to a real system. We apply established ideas from partial identification, minimax regret, randomized decisions, Borda ranking, comparison graphs, and target-first evaluation to candidate-dependent physical contexts.

The closest antecedents already establish interval decisions, graph-based score identifiability and precision, and robust lotteries. Our scoped contribution is their interaction here: an exact candidate-dependent-context construction, its edge-coupled regret values, a route-colored decision program, and an executable finite-sample interface.

2. Pair-first context can leave the global action unidentified

2.1 Target and observed routes

Let $K \geq 3$ policies be indexed by $i = 1, \dots, K$, and let one shared context variable C have the target distribution. For a common-context target, write q_{ij} for policy i 's expected half-credit comparison score against j , with $q_{ji} = 1 - q_{ij}$ and $q_{ii} = 1/2$. This score may be a conditional comparison probability or a deterministic fractional score. The unrestricted construction below treats pair responses as primitive and imposes no shared scalar-outcome, random-utility, stochastic-transitivity, or cross-edge coherence constraint. Under the uniform-reference normalized Borda target,

$$V_i = \frac{\frac{1}{2} + \sum_{j \neq i} q_{ij}}{K}.$$

Borda is the declared comparison target. It equals success-mean ordering in the shared-binary model below, but is not a generic deployment utility.

Now suppose the policy pair is selected before context is constructed. The observed edge for pair $\{i, j\}$ averages outcomes under a pair-specific context mechanism G_{ij} , not necessarily under the common target population.

2.2 Observational-equivalence result

Let the one shared context variable take values A and B with equal target weight. Each pair has a route G_{ij} , degenerate at one of those shared contexts. Fix the complete routed score law as a deterministic half tie. Choose the hidden context-specific primitive pair score so that the target edge can take any value in $[1/4, 3/4]$. These context weights differ from Theorem 3's opponent-reference weights. All completions generate the same observed route/context/score law. Two explicit completions reverse the unique common-context winner.

For every finite $K \geq 3$, one complete-support pair-first half-win law is compatible with the full primitive-response box

$$q_{ij} = \frac{1}{2} + \delta_{ij}, \quad \delta_{ij} \in [-1/4, 1/4],$$

independently for all $\binom{K}{2}$ unordered pairs.

Theorem 1 (construction-bounded non-identification). For every $K \geq 3$, there is a pair-first, complete-pair-support observed law for which all compatible worlds agree on every observed route but the

common-context Borda winner is not identified. Repeating the same routed comparisons without a bridge does not shrink this compatible target box.

Construction and proof: [Supplement S1](#). This counterexample does not apply to every pair-first design.

3. Exact minimax-regret solutions for the constructed edge box

3.1 Uniform opponent reference

Let p be a policy lottery, used here as an ambiguity benchmark rather than a deployment recommendation, chosen before the compatible world and policy draw. Its worst-compatible expected regret is

$$\mathcal{R}(p) = \sup_q \left\{ \max_w V_w - \sum_i p_i V_i \right\}.$$

For a fixed candidate winner w , independent edge extrema yield

$$\mathcal{R}(p) = \frac{1}{4K} \max_w \sum_{i < j} |\{1(i = w) - p_i\} - \{1(j = w) - p_j\}|.$$

Theorem 2 (full-edge uniform-reference minimax). In this compatible box:

- every deterministic singleton has worst-compatible regret $(K - 1)/(2K)$;
- the unique minimax lottery is uniform over all K policies;
- its worst-compatible expected regret is $(K - 1)/(4K)$.

Proof: [Supplement S2](#).

The factor of two concerns ex-ante expected regret. It does not halve the worst regret of the realized policy.

Each policy's marginal interval has width $(K - 1)/(2K)$, which already gives the deterministic regret in Stoye's arbitrary-treatment interval model. An independent product box would give uniform-lottery regret $(K - 1)^2/(2K^2)$. Our smaller value comes from antisymmetric edge coupling: states with one policy at its upper bound and every other policy at its lower bound are infeasible. Supplement S6 gives the comparison.

3.2 Arbitrary nonnegative opponent reference

Let nonnegative reference weights r_i sum to one and define

$$V_i = \sum_j r_j q_{ij}.$$

Order the three smallest weights as $a = r_{(1)} \leq b = r_{(2)} \leq g = r_{(3)}$. The robust objective reduces to

$$\mathcal{R}(p) = \frac{1}{4} \max_w \left[1 - p_w + \sum_{\substack{i < j \\ i, j \neq w}} |r_j p_i - r_i p_j| \right].$$

Theorem 3 (weighted-reference minimax). The exact minimax value is

$$\mathcal{R}^* = \frac{2 - a - b}{8}.$$

The minimizers form the line segment in Supplement S3 and are unique exactly when the second- and third-smallest reference weights are equal.

Corollary 3.1 (hard support exclusion). If the lottery must assign zero mass to zero-reference policies while those policies remain possible oracle comparators, $p = r$ is uniquely minimax with value $1/4$ whenever any reference weight is zero (Supplement S3).

Here r is the opponent-reference distribution, not the pair-sampling distribution or policy lottery p . Its empirical choice lies outside the theorem.

3.3 Shared-success sensitivity

Under a shared binary success outcome $Y_i(C)$, a pair receives one point for a strict success win and one half for a tie. Route every observed pair through shared context A with identical policy outcomes, and let $x_i = E[Y_i(B)] \in [0, 1]$. Then

$$q_{ij} = \frac{1}{2} + \frac{1}{4}(x_i - x_j).$$

The compatible edges form a gradient polytope satisfying $\delta_{ik} = \delta_{ij} + \delta_{jk}$, not the full independent box. Nevertheless, $x = e_1$ and $x = e_K$ give opposite unique common-context winners while preserving the complete observed half-tie law. For any opponent-reference distribution r ,

$$V_i = \frac{1}{2} + \frac{1}{4}(x_i - r^\top x), \quad \mathcal{R}(p) = \frac{1}{4}(1 - \min_i p_i).$$

Thus deterministic worst regret becomes $1/4$, while the unique uniform minimax lottery and value $(K - 1)/(4K)$ are unchanged and r cancels. Theorem 3's weighted optimizer face and Corollary 3.1's uniqueness are therefore response-model dependent. This reflects missing target-context support, not candidate-dependent routing (Supplement S4).

If the observed-context success profile has range $D < 1$, any policy can be uniquely best under a compatible missing-context completion within the shared-success model (Supplement S4).

3.4 Candidate-dependent route graphs

For each positive-target-weight context c , form a route graph whose edges are the pairs observed there. Policy-success means are identified up to one additive offset per component. Connected graphs identify all contextwise differences; disconnected components can retain target-relevant offsets when the outcome bounds leave slack. The exact criterion is zero compatible width for every target difference, not connectivity alone (Supplement S4).

In a three-policy example, pair 1–2 routes through B while 1–3 and 2–3 route through A. The routed law is compatible with unique common-target winners 1 and 3. Its ex-ante minimax lottery is uniquely $(2/3, 0, 1/3)$, with regret $1/12$, rather than the uniform lottery above. The corresponding minimax action is a linear program (Supplement S4).

If a context graph has m components, $m - 1$ cross-component pair types are necessary and sufficient when the allowable-pair quotient graph is connected; with costs, a minimum spanning tree gives the cheapest repair (Supplement S4).

3.4.1 Hypothetical robot design

Suppose policies A, B, C are evaluated in near and far reset strata. Near comparisons $A - C$ and $B - C$ connect the roster, but far contains only $A - B$. Adding either safe $C - A$ or $C - B$ far-stratum comparison closes the gap. The bridge requires comparable randomized trials, recorded policy and reset lineage, and a separate precision analysis.

3.4.2 Finite samples

Replace each exact edge equation by a simultaneous interval, $\ell_c \leq E_c x^c \leq u_c$. Linear programs then give sharp bounds on every target difference, the policies that can still win, any winner certified against all compatible values, and the minimax lottery. If the edge intervals jointly cover with probability $1 - \alpha$, these projected statements inherit that coverage. They do not repair dependence, adaptive sampling, or target drift (Supplement S4.3).

3.4.3 Bradley–Terry structure

Suppose exact common-context comparisons obey $q_{ij} = \sigma(s_i - s_j)$. Applying the inverse link to each observed edge identifies score differences along paths. The Borda-winner set is therefore identified exactly when the comparison graph is connected; a unique winner also requires a unique largest score. Disconnected components retain separate offsets and need bridges. This is an established graph-identifiability consequence, not a new Bradley–Terry result (Supplement S4.4).

3.5 Relation to maximal and robust lotteries

On the observed zero margin matrix, every lottery is maximal. Robust maximal and Borda-regret rules both select the uniform lottery here but use different losses and adversaries (Supplement S5).

4. A different action can remain identified

For each pair $a < b$, define the mechanism-specific edge

$$\theta_{ab}^{pc} = E_{C \sim G_{ab}}[Z_{ab}(C)].$$

If a future unordered pair is drawn from a fixed outcome-independent law π , and the same G_{ab} will be used, a routing rule has value

$$V^{pc}(\delta) = \sum_{a < b} \pi_{ab} [\mathbf{1}\{\delta(a, b) = a\} \theta_{ab}^{pc} + \mathbf{1}\{\delta(a, b) = b\} (1 - \theta_{ab}^{pc})].$$

The objective is separable, so the optimal same-mechanism within-pair choice selects the locally preferred member of each positive-weight pair.

In a three-edge example, the local rule has value 3/4, while always choosing the lower-index policy has value 7/12, even though all tournament scores tie at 1/2. This target requires a fixed future pair law, positive edge support, stable pair mechanisms, and assignment-aware uncertainty (Supplement S5).

5. What public records can identify

We fixed three public releases before aggregate analysis. They are purposive examples, and missing public fields do not imply private absence.

5.1 A positive finite-panel case

Three AnkIle R5 releases each run three fixed policy artifacts on the same 50 declared Sobol states: routing, marker, and square ([routing](#); [marker](#); [square](#)). All 450 policy-state cells are present. The success counts are ((8,9,12)), ((19,32,31)), and ((28,36,33)), so the exact released-panel winner is identified in each task, by margins of 3, 1, and 3 successes. Each three-edge route graph is complete.

This identifies only the displayed states. One rollout per cell supplies no population uncertainty law, and the files do not reconstruct achieved reset acceptance or the full retry ledger. The omission can matter at

the scale of the reported differences. Under a worst-case sensitivity that replaces an entire retained three-policy round by another binary outcome vector, reversing the released winner requires only two routing rounds, one marker round, or two square rounds. This is not evidence that any such replacement occurred. The released configuration enables incomplete-round reruns for routing and marker, but not square, so the retained attempt history is needed to distinguish a stable result from selection through reruns.

5.2 Pair support is not a common-context graph

The pinned RoboArena release contains 3,883 sessions, 21 policy labels, and 104 of 210 possible label pairs ([data](#); [paper](#)). One unpaired label forms a separate component; the other 20 labels are connected. Twenty-seven observed edges appear in only one session.

This is policy-pair support, not the contextwise route graph of Section 3.4. Tasks and scenes are constructed for the selected pair, and the release does not bind assignment weights or pool epochs, exact initial states and resets, robot instances, or retry ancestry to a common target. Connectivity among the 20 compared labels therefore does not identify a common-context winner.

5.3 A strong protocol can lose its join at release

TRI reports 1,800 real-world rollouts in blind, randomized test bundles that hold the initial condition across compared policies ([paper](#); [data](#)). Dryad version 4 releases outcome arrays, but no bundle, rollout, initial-condition, realized-order, reset, session, robot, retry, or immutable policy-version identifier. Two published success rates also disagree with their own arrays and counts. The public tables support marginal checks, not reconstruction of the matched route edges.

Supplement S7 compares five additional public systems using the same evidence layers.

6. What the missing records must answer

For a finite common-state policy comparison, the practical questions are simple: were the declared states actually achieved comparably for every policy, which attempts were retained or replaced, and could that choice depend on policy or observed performance? An attempt record needs only enough fields to join state, policy and realized order to reset status, execution status, outcome, and any replacement. Video, operator identity, and fine-grained telemetry are not required for this check.

For AnkIle, the decisive reanalysis is to compare the published retained panel with the first valid attempt per policy-state cell and with all valid attempts, averaged within each cell before restoring the original equal weighting of the 50 states. This prevents frequently retried cells from receiving extra weight. If the winner is stable, the provenance strengthens the finite-panel claim. If it changes, the size and direction of the retention effect should be reported. For a disconnected route graph, the substantive repair remains a safe common-context bridge; repeating an existing route cannot identify the missing comparison.

7. Limitations

These counterexamples do not characterize every pair-first design or recommend a policy lottery. Theorems 2-3 use a primitive-response box; the shared-success sensitivity uses a narrower model. The real records are purposive. AnkIle identifies only three exact released panels; its thin winner margins are sensitive to one or two worst-case matched-round replacements, and the public files do not show whether such replacements occurred. RoboArena supplies pair support but not a common-context target, and TRI's public tables do not retain its matched-bundle join.

The finite-sample program inherits the validity of supplied simultaneous edge intervals. It does not provide a sampling unit, power, minimum detectable effect, or stopping rule; each bridge study still needs those design choices.

The English-language review does not claim priority for the underlying methods.

Data and code availability

Data, code, and proofs are available in the package and supplement. Project materials: <https://github.com/trilamsr/research/tree/main/sim2real-correlation-audit>.

Author and declarations

Tri Lam developed the analysis, conducted the public-record review, and wrote the manuscript. No conflicts of interest are declared. The paper reports no new human- or animal-subject experiment.

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